Compressor off-design report: summary and diagnostics

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Rotor-Stator  |  run: Max_PR_Compressor_design_rs_with_last_stator

Rotor-Stator  |  PR_des = 1.8264, eta_des = 0.7804  |  correlations: polynomial, Howell factor 2.00

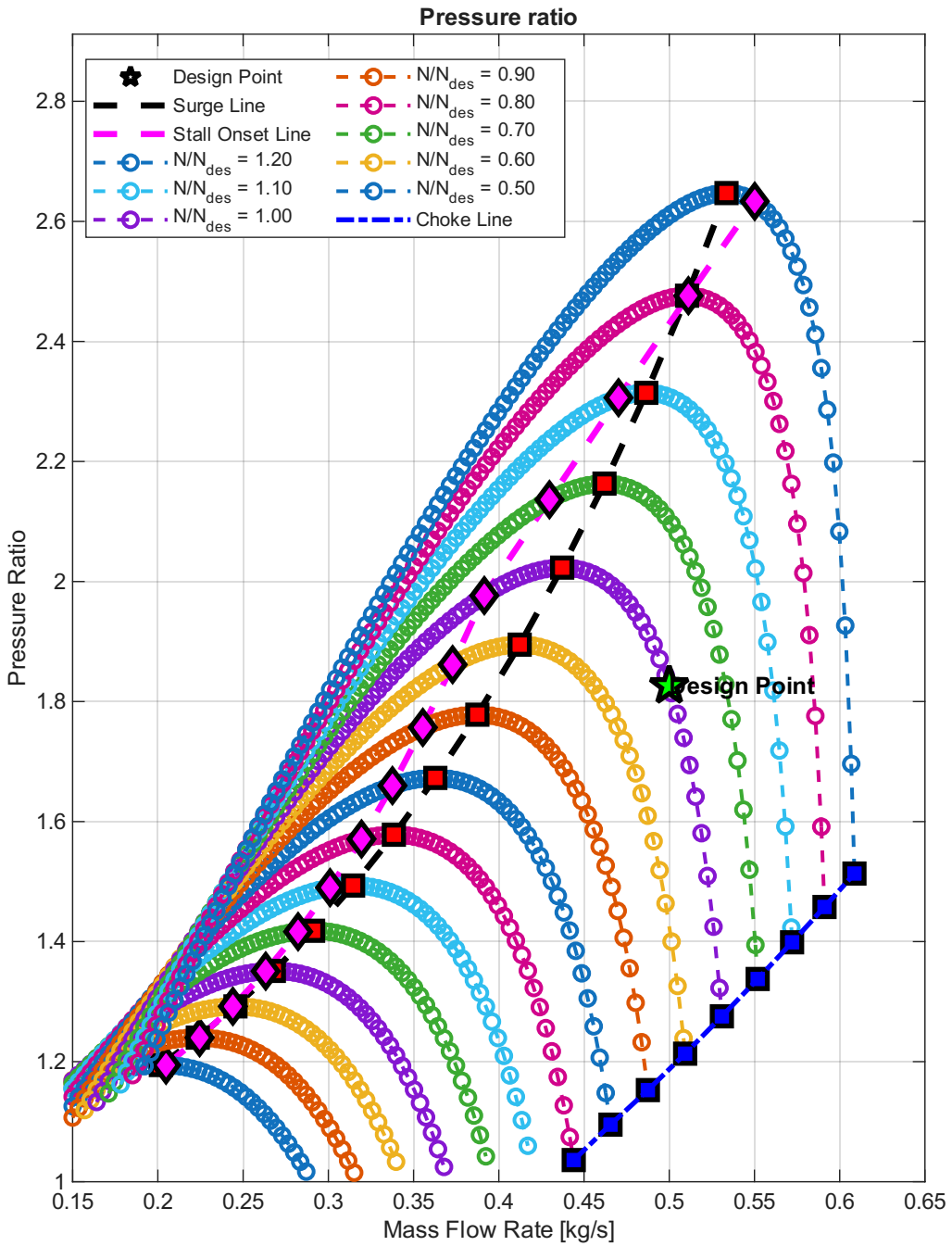
STABILITY (design speed): first limit = peak at m = 0.4372 kg/s (PR 2.023); stall margin SM = 26.7 %
   peak of the speed line (zero slope): m = 0.4372 kg/s | Howell stall onset: m = 0.3915 kg/s
   Koch stage limit: not reached on the computed part of the line
CHOKE (design speed): m_choke = 0.5309 kg/s (Stator 2 exit (absolute)) -> choke margin 6.2 %, operating range (m_choke - m_lim)/m_choke = 17.6 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.406 -> surge if the installation has V_p/(A_c L_c) > 3.88 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Stator 2 (omega/omega_des = 1.57), lowest de Haller Stator 2 (0.55). Details: Tables > Blade
      rows.

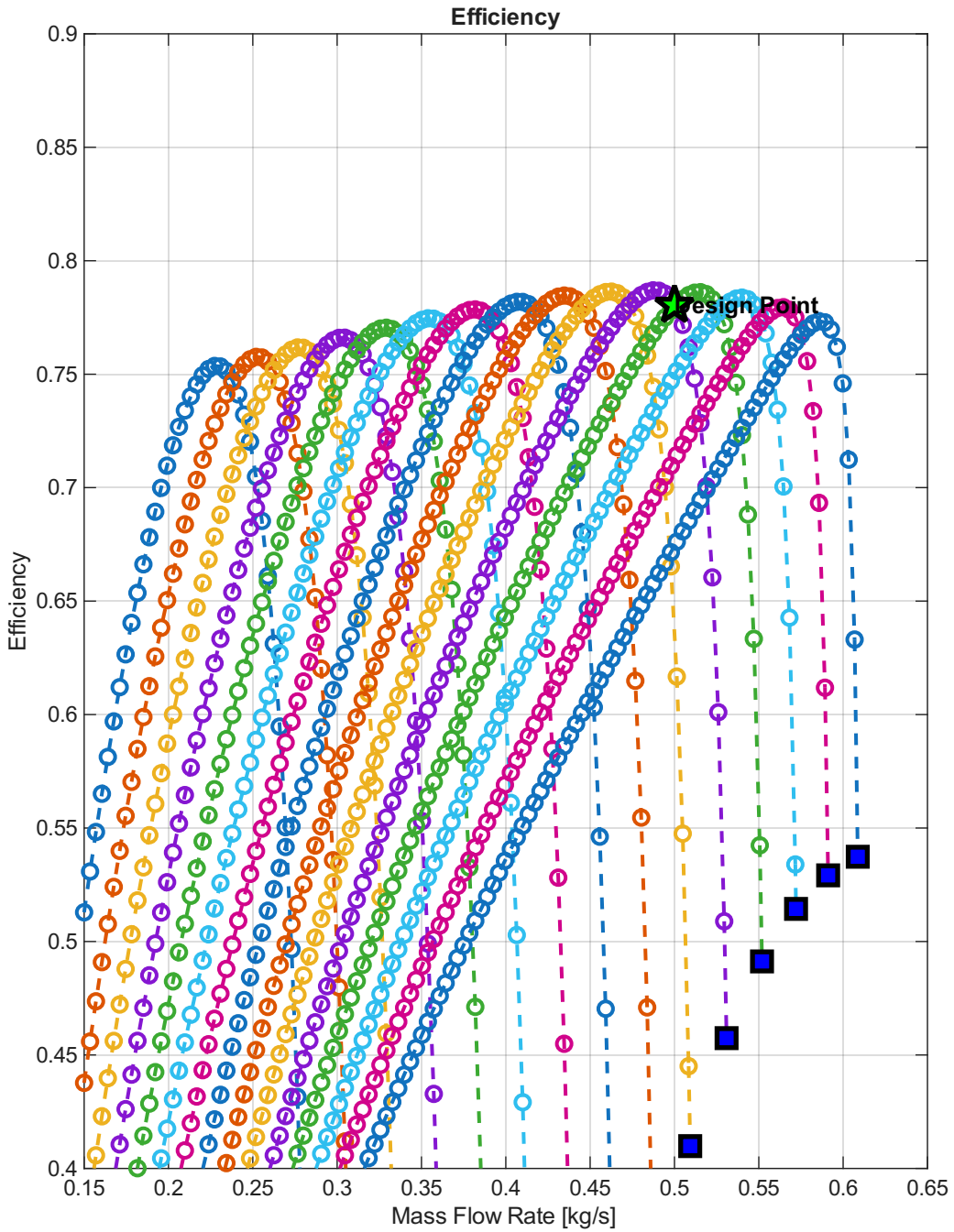
KOCH STAGE-LOADING INDICATOR (never a stability limit): INDICATIVE
   Indicative: near Koch's data. Read the ratio as a trend (more or less loaded than a typical stage), not as a limit. Howell and the peak set the
      limit.
   Koch line: none to draw. No speed line reaches Ch_ef/Ch_max = 1 before the peak or Howell limit, so the stage stays below its Koch limit over the
      whole stable range.
   why: Stage 1 (R1+S1): L/g2 = 2.41 is above 2.2 (high solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: Stage 2 (R2+S2): L/g2 = 2.41 is above 2.2 (high solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Stage 1 (R1+S1): OK (Ch_ef/Ch_max = 0.73 <= target 0.85)
     - margin available: the solidity could be LOWERED x0.66 (stage L/g2 2.41 -> 1.60): R1 1.97 -> 1.31, S1 1.97 -> 1.31; re-run the design script
       with it (profile loss changes)
     - or RAISE the stage static-pressure rise by up to about 16% before reaching the target
     - note: outside Koch's data: L/g2 = 2.41 (data 0.6-2.2) -> indicative only
  Stage 2 (R2+S2): OK (Ch_ef/Ch_max = 0.73 <= target 0.85)
     - margin available: the solidity could be LOWERED x0.66 (stage L/g2 2.41 -> 1.58): R2 1.97 -> 1.29, S2 1.97 -> 1.29; re-run the design script
       with it (profile loss changes)
     - or RAISE the stage static-pressure rise by up to about 17% before reaching the target
     - note: outside Koch's data: L/g2 = 2.41 (data 0.6-2.2) -> indicative only
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Inputs of this run (.mat data and settings)

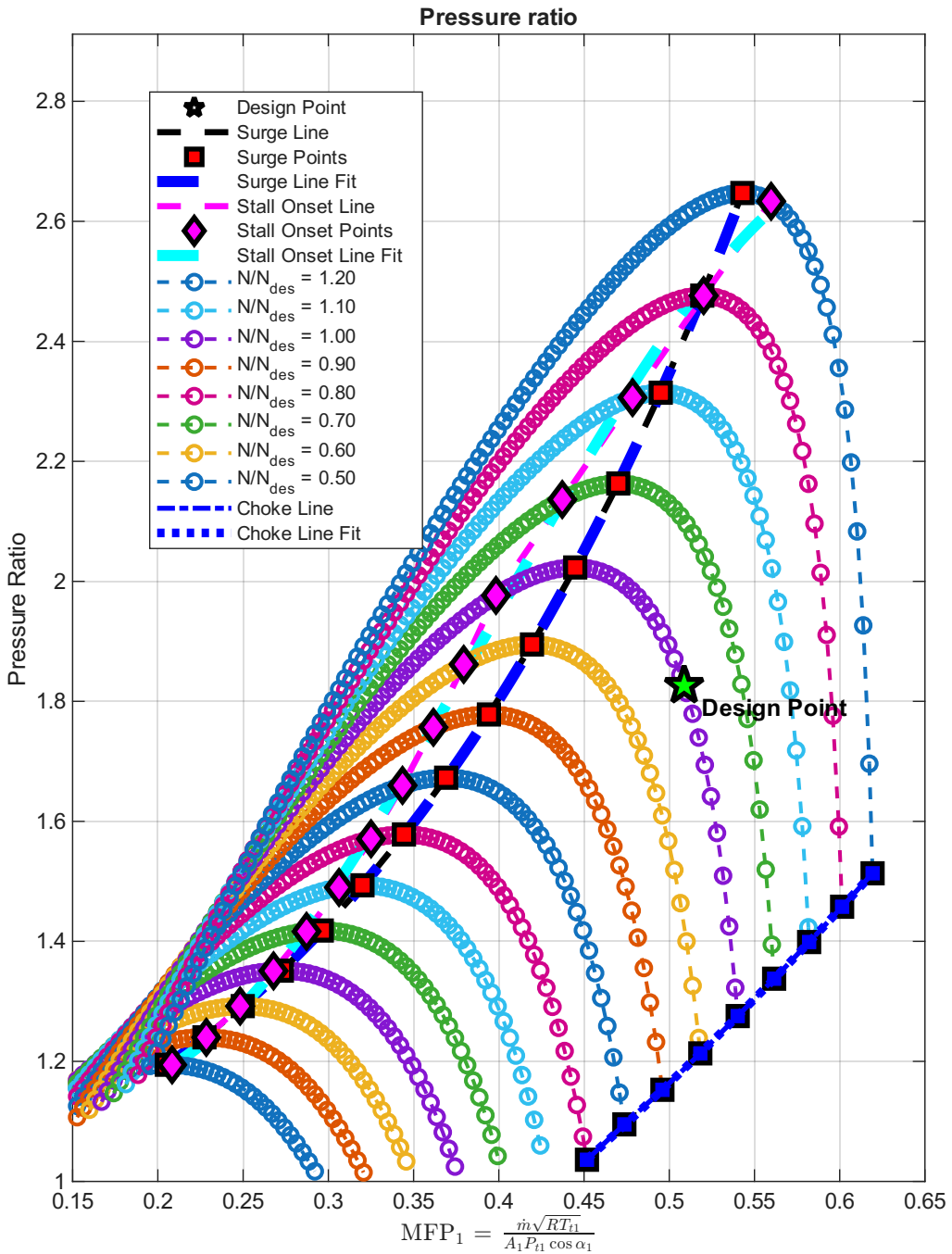
Variable	Value	Unit	Source
Configuration	Rotor-Stator		
Input source	.mat: Max_PR_Compressor_design_rs_w...		
alpha1	38.4033	deg	.mat: Max_PR_Compressor_design_rs_w...
n_stages	2	-	.mat: Max_PR_Compressor_design_rs_w...
A_vec	0.00360564 0.00323586 0.00290087 0....	m^2	.mat: Max_PR_Compressor_design_rs_w...
beta1_dash	51.8472	deg	.mat: Max_PR_Compressor_design_rs_w...
beta2_dash	34.8686	deg	.mat: Max_PR_Compressor_design_rs_w...
m_dot	0.5	kg/s	.mat: Max_PR_Compressor_design_rs_w...
w_loss_rotor	0.140799	-	.mat: Max_PR_Compressor_design_rs_w...
deviation	3.53466	deg	.mat: Max_PR_Compressor_design_rs_w...
incidence	6.87796	deg	.mat: Max_PR_Compressor_design_rs_w...
deflection	20.3219	deg	.mat: Max_PR_Compressor_design_rs_w...
w_loss_stator	0.115656	-	.mat: Max_PR_Compressor_design_rs_w...
U	315.356	m/s	.mat: Max_PR_Compressor_design_rs_w...
sigma_rotor	1.97032	-	.mat: Max_PR_Compressor_design_rs_w...
sigma_stator	1.97032	-	.mat: Max_PR_Compressor_design_rs_w...
chord	0.01	m	.mat: Max_PR_Compressor_design_rs_w...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	135	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

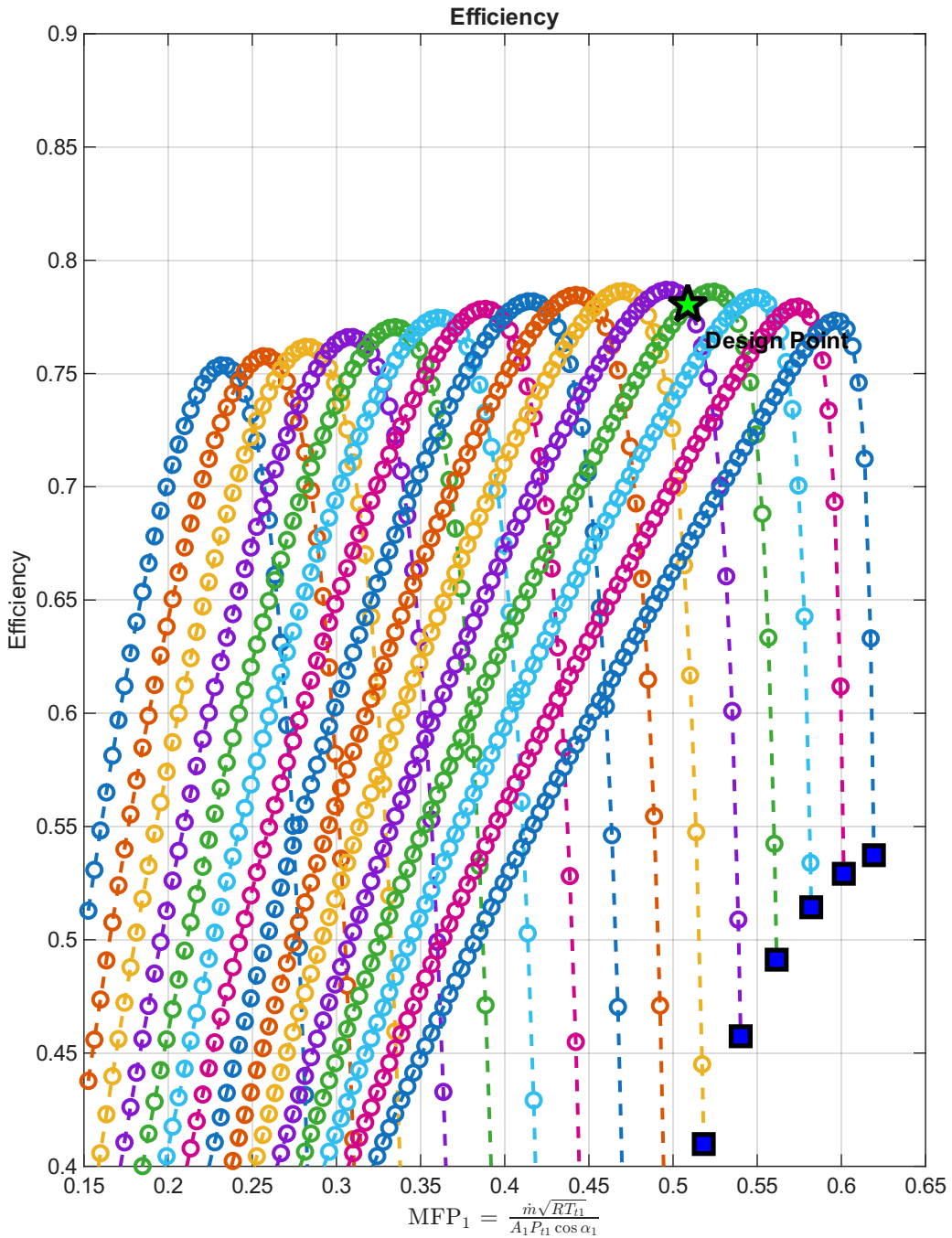
Standard (dimensional) map - Rotor-Stator



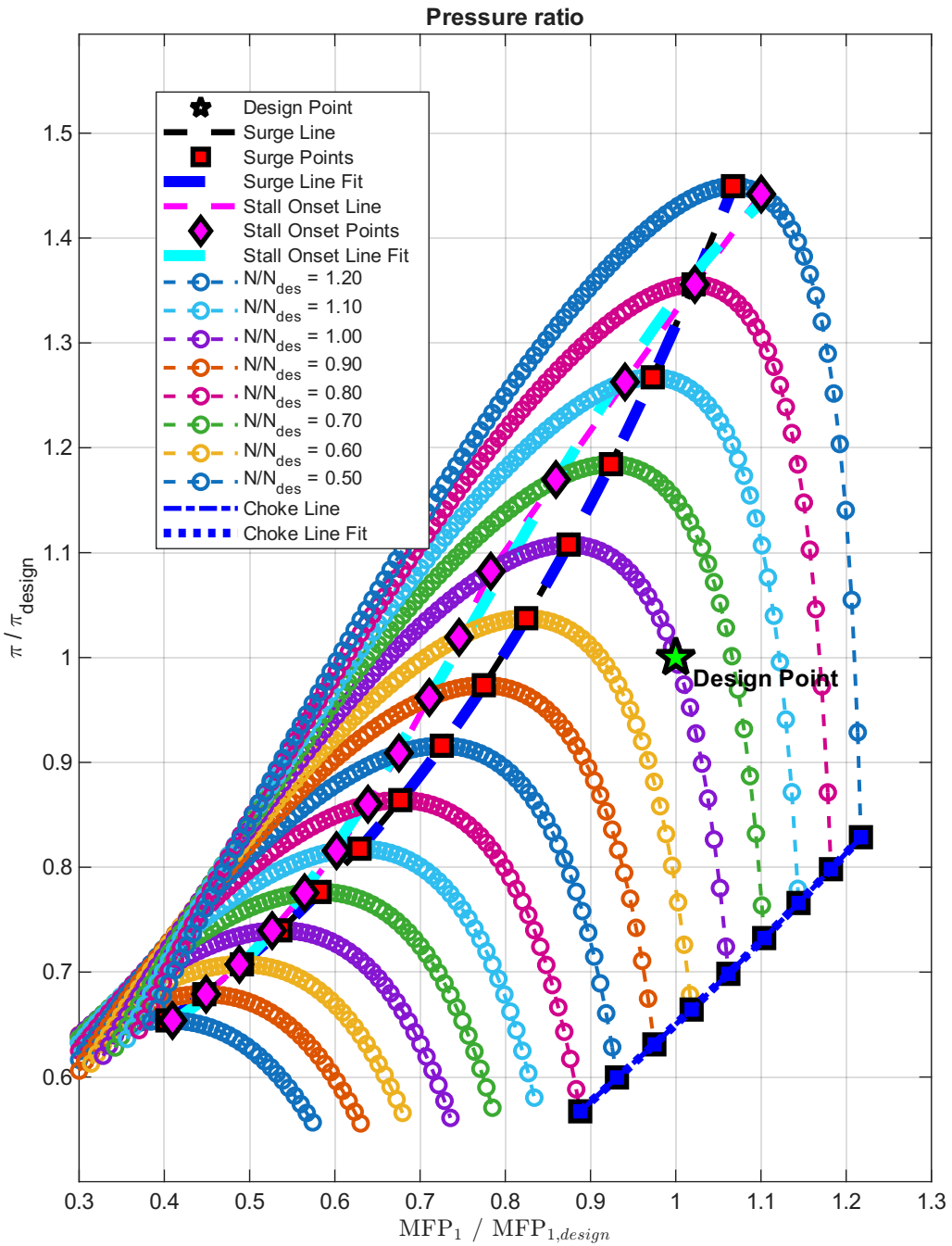


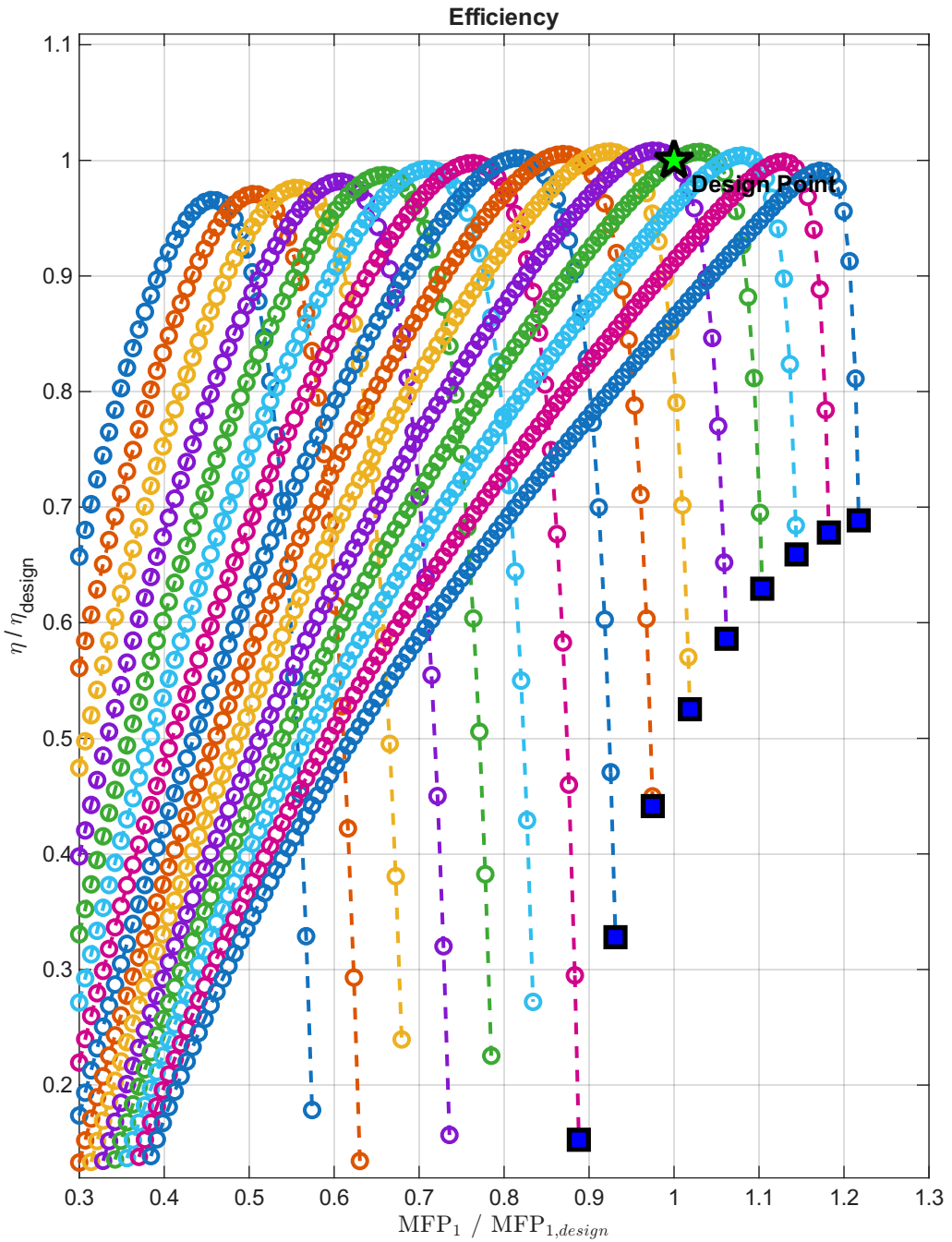
Non-dimensional map - Rotor-Stator



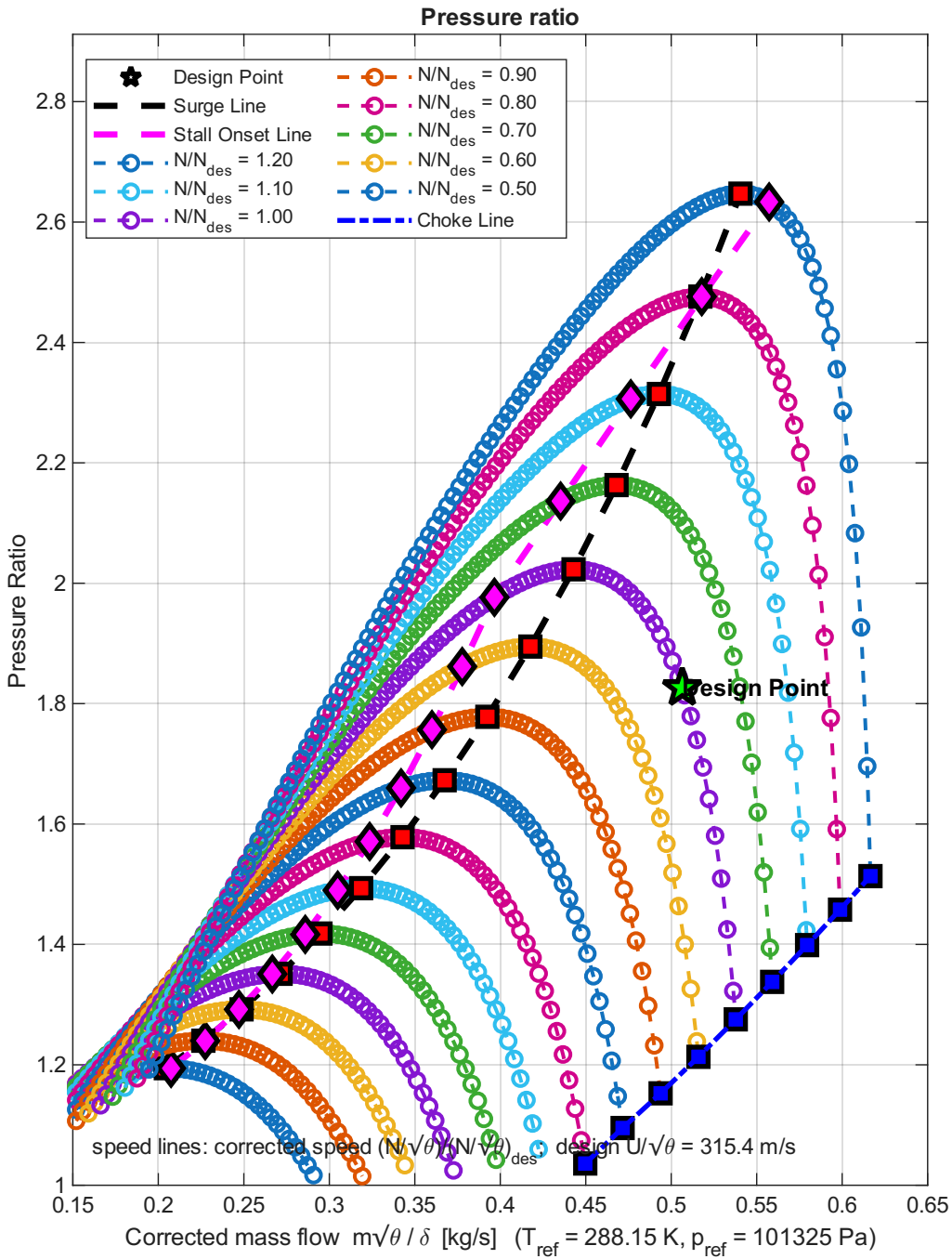


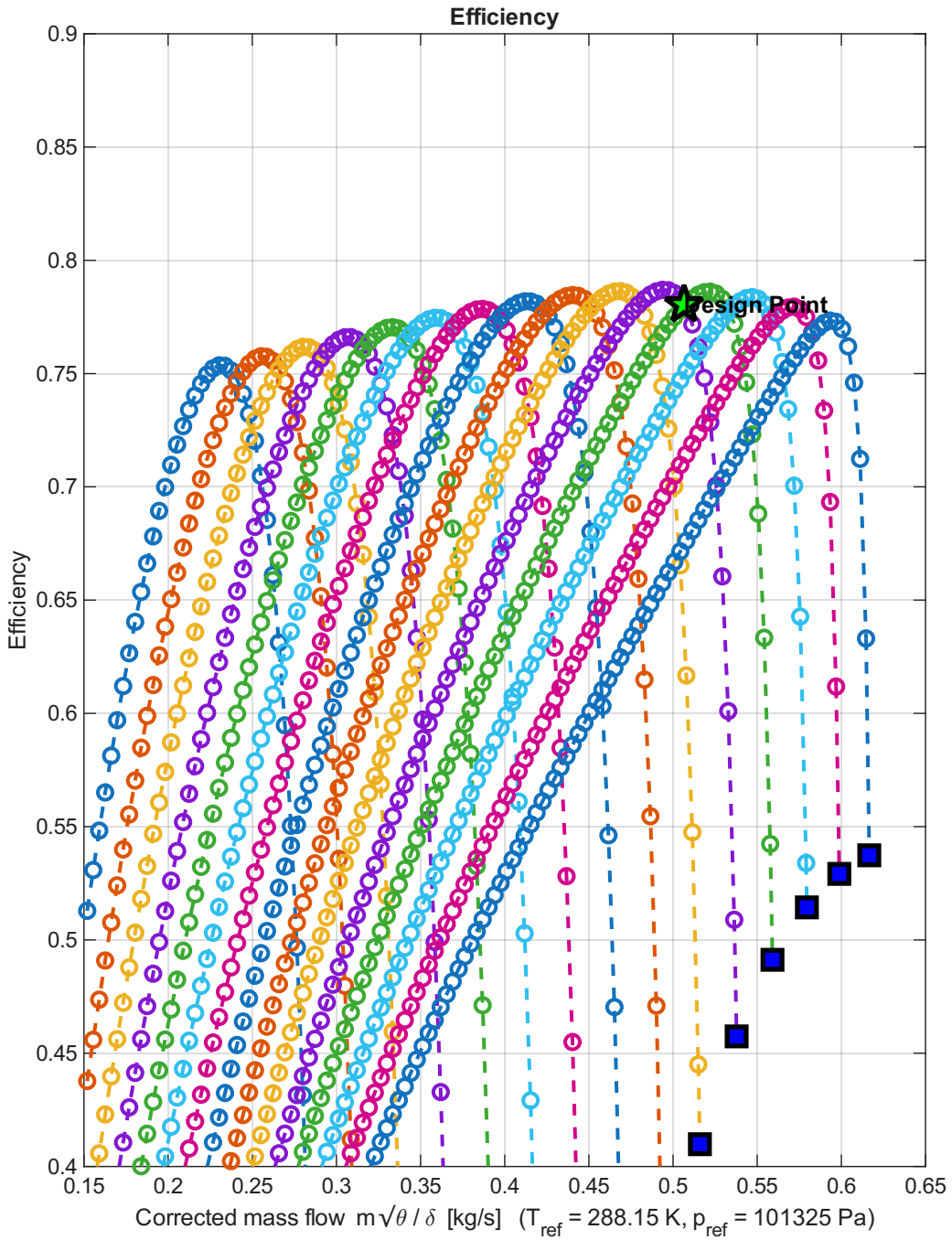
Relative non-dimensional map - Rotor-Stator





Corrected map - Rotor-Stator





Fitted boundary equations (surge, stall onset, choke)

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==== Non-dimensional Compressor Map: Pressure Ratio - Rotor-Stator ====
Surge line:
Surge_PR = 7.7662*MFP^3 + -0.7344*MFP^2 + 1.3691*MFP + 0.8759   [R^2 = 1.00000]
Stall-onset line (Howell):
StallOnset_PR = -30.9459*MFP^3 + 36.8489*MFP^2 + -9.6010*MFP + 1.8740   [R^2 = 0.99899]
Choke line:
Choke_PR = 1.8115*MFP^2 + 0.9036*MFP + 0.2590   [R^2 = 0.99998]
==== Relative Non-dimensional Compressor Map: Pressure Ratio - Rotor-Stator ====
Surge line:
(PR/PR_design) = 0.5599*x^3 + -0.1041*x^2 + 0.3814*x + 0.4796   (x = MFP/MFP_design)   [R^2 = 1.00000]
Stall-onset line (Howell):
(PR/PR_design) = -2.2310*x^3 + 5.2219*x^2 + -2.6744*x + 1.0261   (x = MFP/MFP_design)   [R^2 = 0.99899]
Choke line:
(PR/PR_design) = 0.2567*x^2 + 0.2517*x + 0.1418   (x = MFP/MFP_design)   [R^2 = 0.99998]
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Design point

Quantity	Value
Configuration	Rotor-Stator
Pressure ratio PR_des	1.8264
Isentropic efficiency eta_des	0.7804
Mass flow m_des [kg/s]	0.5
MFP_1,des	0.50874
Corrected mass flow [kg/s]	0.50649
Inlet Tt1 [K] / Pt1 [Pa]	288 / 100000
Correlation set	polynomial
Howell factor	2

Choke

N/N_des	m_choke [kg/s]	MFP_1	PR	Eff	Choking row	Inside compressor region (PR>1)
0.5	0.32657	0.3323	0.732		Stator 2 exit (absolute)	no
0.55	0.34421	0.3502	0.776		Stator 2 exit (absolute)	no
0.6	0.36267	0.369	0.822		Stator 2 exit (absolute)	no
0.65	0.38225	0.3889	0.873		Stator 2 exit (absolute)	no
0.7	0.40226	0.4093	0.925		Stator 2 exit (absolute)	no
0.75	0.42284	0.4302	0.979		Stator 2 exit (absolute)	no
0.8	0.44397	0.4517	1.036	0.119	Stator 2 exit (absolute)	yes
0.85	0.46554	0.4737	1.095	0.256	Stator 2 exit (absolute)	yes
0.9	0.48748	0.496	1.153	0.344	Stator 2 exit (absolute)	yes
0.95	0.50933	0.5182	1.213	0.41	Stator 2 exit (absolute)	yes
1	0.53089	0.5402	1.275	0.457	Stator 2 exit (absolute)	yes
1.05	0.55191	0.5616	1.338	0.491	Stator 2 exit (absolute)	yes
1.1	0.57208	0.5821	1.399	0.514	Stator 2 exit (absolute)	yes
1.15	0.5911	0.6014	1.458	0.529	Stator 2 exit (absolute)	yes
1.2	0.60867	0.6193	1.513	0.537	Stator 2 exit (absolute)	yes

Stall onset (Howell)

N/N_des	m_stall [kg/s]	PR	Eff	Stalling row
0.5	0.20478	1.194	0.724	Rotor 1
0.55	0.22449	1.24	0.722	Rotor 1
0.6	0.24399	1.292	0.72	Rotor 1
0.65	0.26326	1.351	0.717	Rotor 1
0.7	0.28227	1.417	0.713	Rotor 1
0.75	0.30102	1.49	0.708	Rotor 1
0.8	0.31947	1.571	0.702	Rotor 1
0.85	0.33762	1.66	0.695	Rotor 1
0.9	0.35544	1.757	0.687	Rotor 1
0.95	0.37292	1.862	0.677	Rotor 1
1	0.39152	1.977	0.669	Stator 2
1.05	0.42973	2.136	0.69	Stator 2
1.1	0.47027	2.306	0.71	Stator 2
1.15	0.51125	2.476	0.727	Stator 2
1.2	0.55016	2.633	0.739	Stator 2

Peak (zero slope)

N/N_des	m_peak [kg/s]	PR	Eff	m_stall - m_peak [kg/s]	First limit on throttling
0.5	0.20176	1.194	0.716	0.00302	stall onset (Howell)
0.55	0.22342	1.24	0.72	0.00108	stall onset (Howell)
0.6	0.2455	1.292	0.723	-0.00151	peak (zero slope)
0.65	0.26805	1.351	0.727	-0.00479	peak (zero slope)
0.7	0.29106	1.418	0.73	-0.00878	peak (zero slope)
0.75	0.31453	1.493	0.734	-0.01352	peak (zero slope)
0.8	0.33846	1.578	0.736	-0.01899	peak (zero slope)
0.85	0.36278	1.673	0.739	-0.02517	peak (zero slope)
0.9	0.38743	1.778	0.74	-0.03199	peak (zero slope)
0.95	0.41228	1.895	0.74	-0.03937	peak (zero slope)
1	0.43721	2.023	0.739	-0.04569	peak (zero slope)
1.05	0.46202	2.163	0.737	-0.03229	peak (zero slope)
1.1	0.48652	2.314	0.732	-0.01626	peak (zero slope)
1.15	0.51047	2.476	0.726	0.00078	stall onset (Howell)
1.2	0.53361	2.647	0.718	0.01656	stall onset (Howell)

Koch limit

N/N_des	m_Koch [kg/s]	PR	Eff	Limiting stage
0.5	NaN	NaN	NaN	not reached
0.55	NaN	NaN	NaN	not reached
0.6	NaN	NaN	NaN	not reached
0.65	NaN	NaN	NaN	not reached
0.7	NaN	NaN	NaN	not reached
0.75	NaN	NaN	NaN	not reached
0.8	NaN	NaN	NaN	not reached
0.85	NaN	NaN	NaN	not reached
0.9	NaN	NaN	NaN	not reached
0.95	NaN	NaN	NaN	not reached
1	NaN	NaN	NaN	not reached
1.05	NaN	NaN	NaN	not reached
1.1	NaN	NaN	NaN	not reached
1.15	NaN	NaN	NaN	not reached
1.2	NaN	NaN	NaN	not reached

Stability limit & Greitzer B

N/N_des	First limit	m_lim [kg/s]	PR	U [m/s]	a_p [m/s]	U/(2a_p)	(V_p/(A_c L_c))_crit	B	Beyond the limit
0.5	stall onset	0.20478	1.194	157.7	352.2	0.2239	12.77	NaN	-
0.55	stall onset	0.22449	1.24	173.4	354.8	0.2444	10.71	NaN	-
0.6	peak	0.2455	1.292	189.2	357.6	0.2646	9.14	NaN	-
0.65	peak	0.26805	1.351	205	360.6	0.2842	7.92	NaN	-
0.7	peak	0.29106	1.418	220.7	363.8	0.3034	6.95	NaN	-
0.75	peak	0.31453	1.493	236.5	367.2	0.322	6.17	NaN	-
0.8	peak	0.33846	1.578	252.3	370.9	0.3401	5.53	NaN	-
0.85	peak	0.36278	1.673	268.1	374.9	0.3575	5.01	NaN	-
0.9	peak	0.38743	1.778	283.8	379	0.3744	4.57	NaN	-
0.95	peak	0.41228	1.895	299.6	383.4	0.3906	4.19	NaN	-
1	peak	0.43721	2.023	315.4	388.1	0.4063	3.88	NaN	-
1.05	peak	0.46202	2.163	331.1	393	0.4213	3.61	NaN	-
1.1	peak	0.48652	2.314	346.9	398.2	0.4356	3.37	NaN	-
1.15	stall onset	0.51125	2.476	362.7	403.4	0.4495	3.17	NaN	-
1.2	stall onset	0.55016	2.633	378.4	407	0.465	2.96	NaN	-

Blade rows

Operating point (N = 1)	m [kg/s]	Blade row	Incidence i [deg]	(i - i_des)/deflection	Loss omega	omega/omega_des	M_in (row frame)	M_out (row frame)	de Haller V_out/V_in	Deviation [deg]	Watch
Design point	0.5	Rotor 1	6.88	0	0.1408	1	0.75	0.482	0.662	3.53	de Haller < 0.72
Design point	0.5	Stator 1	6.88	0	0.1157	1	0.727	0.468	0.662	3.53	de Haller < 0.72
Design point	0.5	Rotor 2	6.88	0	0.1408	1	0.707	0.456	0.662	3.53	de Haller < 0.72
Design point	0.5	Stator 2	6.88	0	0.1157	1	0.688	0.444	0.662	3.53	de Haller < 0.72
First limit (peak)	0.43721	Rotor 1	12.65	0.284	0.1911	1.357	0.758	0.401	0.55	5.21	de Haller < 0.72
First limit (peak)	0.43721	Stator 1	12.82	0.292	0.169	1.461	0.718	0.383	0.553	5.3	de Haller < 0.72
First limit (peak)	0.43721	Rotor 2	13.2	0.311	0.2013	1.429	0.694	0.368	0.548	5.52	de Haller < 0.72
First limit (peak)	0.43721	Stator 2	13.45	0.323	0.1811	1.566	0.671	0.355	0.546	5.67	de Haller < 0.72
Near choke (0.99 m_choke)	0.52558	Rotor 1	4.22	-0.131	0.1514	1.075	0.749	0.536	0.733	3.45	
Near choke (0.99 m_choke)	0.52558	Stator 1	3.03	-0.19	0.1377	1.19	0.73	0.548	0.766	3.47	
Near choke (0.99 m_choke)	0.52558	Rotor 2	1.14	-0.282	0.189	1.343	0.713	0.579	0.825	3.52	
Near choke (0.99 m_choke)	0.52558	Stator 2	-2.15	-0.444	0.2333	2.017	0.702	0.664	0.951	3.51	